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Source: *The Journal of Finance*, Dec., 1983, Vol. 38, No. 5 (Dec., 1983), pp. 1543-1568

Published by: Wiley for the American Finance Association

Stable URL: <http://www.jstor.com/stable/2327586>

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On the Positive Role of Financial Intermediation in Allocation of Venture Capital in a Market with Imperfect Information

YUK-SHEE CHAN*

ABSTRACT

This paper develops a theory of financial intermediation that highlights the contribution of intermediaries as informed agents in a market with imperfect information. We consider a venture capital market where the entrepreneurs select the qualities of projects and their perquisite consumptions, about which the investors are imperfectly informed. It is shown that when all investors have positive search costs, the entrepreneurs are induced to offer the unacceptable inferior projects ("lemons" only), and the investors will not enter the venture capital market, but put their funds in other low return investments—an undesirable allocation of resources.

Beginning with an initial undesirable situation, the financial intermediaries may evolve as informed agents that induce a Pareto-preferred allocation, leading the investors to a higher welfare state. We focus our analysis on the existence of intermediation equilibria when the market for intermediation services is competitive. The distribution of returns on projects, the fees charged by intermediaries, and the fraction of institutional holdings are all endogenous in equilibrium. It is shown that (i) there cannot be a competitive intermediation equilibrium with very high institutional holdings, and (ii) in other cases multiple equilibria may exist, but the one with the highest institutional holdings dominates the others in a Pareto sense.

THIS PAPER IS MOTIVATED by interesting and important recent research in financial intermediation from the perspective of imperfect information (e.g., Leland and Pyle (1977) and Campbell and Kracaw (1980)). In these studies, the intermediaries serve as information production agents about the qualities of firms.

As is well known in the information literature, there may not be a market for information because of appropriability problems and because of the reliability of the information provided. In their work using a signaling model, Leland and Pyle have rationalized that the existence of financial intermediaries can solve the above problems in markets with imperfect information. Their observation is important because it provides a foundation for further work on the contribution of financial intermediation to the welfare of investors. Indeed, a survey of the

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literature on financial intermediation (see Section III A) reveals that much of the work has not been able to explicitly identify a positive role to the intermediaries. For example, Campbell and Kracaw's (1980) intermediary distinguishes good firms from the bad, and as a result the firms' true values are reflected in the market, whereas without information production all firms are priced at their average value. The net result is a wealth redistribution from the owners of the bad firms to those of the good firms, which may serve the purposes of "equity" and increase the informational efficiency of the market. However, the aggregate wealth of the owners of the firms has not changed and in fact it has decreased net of the information costs borne by owners. The reason behind this is that the qualities of firms are given exogenously and they are not affected by the information acquisition process. This parallels the well-known finding that when information induces no changes in production, it may have no social value (see Hirshleifer (1971) and Hakansson et al. (1982)). The implication is unfortunate—the information acquisition activities of financial intermediation make no positive contribution. We think such implication is inaccurate, because we expect an interaction between the asset market that allocates investable resources and the market for information (see Stiglitz (1975)).

Although the effects of information acquisition on the production decisions of economic agents has received little attention in the finance literature, numerous studies have addressed the question of information effects in resource allocation in the context of goods and service markets (e.g., see Akerlof (1970), Salop and Stiglitz (1977), Chan and Leland (1982, 1983), and references therein). For instance, Chan and Leland have shown that firms are induced to offer better price/quality combinations in a more informed market. We consider a similar model of the venture capital market to see whether some of the earlier results are robust in a financial market context and to draw out the implications of the model.

In our model, an entrepreneur (or venture capitalist) selects the quality of a project (which depends on the time and effort expended on the project) and his perquisite consumption diverted from the returns on the project. Although investors may have rational expectation regarding the entrepreneurial actions in equilibrium, the actions are unobserved by investors. However, investors can learn about the entrepreneurs through search, which can be done at a cost that may vary across investors. Specifically, we show that the perquisite/quality combinations that are selected by the entrepreneurs (which determine investor returns) will depend on the distribution of the search costs. The presence of some zero-cost or perfectly informed investors in the market induces entrepreneurs to select projects with higher investor returns. Without zero-cost investors, the entrepreneurs will undertake only these projects offering low returns and investors will not participate in the venture capital market. This result can be contrasted with that of Jensen and Meckling (1976) where an (agency) cost emerges because the managers cannot convince investors that they will restrict their perquisite consumption. In our model, when all investors are uninformed, the entrepreneurs are induced to undertake inferior projects and investors will not enter the market. Given that informed investors affect the allocation of resources and the welfare of the investors, we address the question of whether

some policies (or institutions) can lead the economy to a higher welfare state, net of costs of such policies, if all investors have search costs.

We explain that, since the intermediaries' costs are paid by clients, the intermediaries play the role of zero-cost, informed agents in the market, who increase the welfare of investors by inducing the entrepreneurs to offer higher return projects. Further, as information about firms is reusable for different clients, there are economies of scale in the intermediary's operations, even if the information cost is assumed to be the same for investors and intermediaries. The feasibility of our intermediaries can be contrasted with that of Diamond (1982). Diamond's intermediary adds a second layer of delegation between the project managers and the ultimate investors. Even though Diamond assumes that the monitoring costs are the same for the intermediary (on managers) and the investors (on intermediary), he demonstrates that intermediation is valuable because of diversification effects. It is important to note that whereas Diamond is concerned with the moral hazard problem with intermediation, we focus on the contribution of intermediaries as informed agents to resource allocation and welfare of investors.

We consider the equilibrium with competitive intermediation. The distribution of returns on projects, the fees charged by intermediaries, and the fraction of institutional holdings (the intermediaries' scale of operation) are all endogenous in equilibrium. We show that a competitive intermediation equilibrium will not exist when too many investors seek to put funds in the intermediaries or when no one seeks to put funds in them. This result parallels that of Grossman and Stiglitz (1980) that informationally efficient markets are untenable. We also show that there may be multiple competitive intermediation equilibria which can be ranked by the Pareto criterion with the fraction of institutional holdings as the measure of rank.

The paper is organized as follows. In Section I we describe the entrepreneurs, the investors, and their behavior. In Section II we define the equilibrium and relate the equilibrium distribution of perquisite/quality combinations to the distribution of investors' search costs. Section III analyzes the competitive intermediation equilibrium and presents the results. Section IV concludes.

I. The Model

A. Entrepreneurs

Consider a market with many entrepreneurs (venture capitalists) trying to raise funds for the projects. The entrepreneurs have access to the same technological information. The "quality" of the project (defined in the next paragraph) is determined by entrepreneur's effort, e . The quality of the project will be $q(e)$ which is assumed to be increasing and concave in e . In the following, we shall consider q as the entrepreneur's choice variable and $e(q)$ will be the effort required to develop a project of quality q ($e(0) = 0$). Thus, $e(q)$ is increasing and convex in q .

We consider a simple one-period model. A project of quality q will generate a random return $\tilde{X}(q)$ per dollar of investment. If the investment is I , then the

total return is $I \tilde{X}(q)$. We assume $I \leq S$, where S is the maximum investment per project which is the same for all ventures. Given that an entrepreneur has made an effort $e(q)$ to develop a project of quality q , there are returns to scale for all $I < S$. As we shall see later, in a market with costly information, such returns to scale may not be fully exploited. For simplicity and to highlight the problem of quality choices with imperfect information, we assume all risks are project-specific and diversifiable. Let $\bar{X}(q)$ be the expected value of $\tilde{X}(q)$ and assume that $\bar{X}$ is increasing and concave in q ($\bar{X}(0) = 0$), so that there is diminishing return to quality. Aside from the selection of the project quality, the entrepreneurs also determine the fraction, p , to consume as perquisites from the returns to the projects, which compensate them for their effort. All entrepreneurs are assumed to be well-diversified and they have the same preference function depending on their effort and their expected perquisite consumption, $\bar{C}$. The entrepreneurs' preference is given by the function $M(\bar{C}, e)$ with $M_1 > 0$ and $M_2 < 0$. If an entrepreneur chooses to develop a project of quality q and to consume a fraction of returns p as perquisites, and if the total investment is I , the entrepreneurs utility is $M(pI \bar{X}(q), e(q))$. An inactive entrepreneur obtains $M(0, 0) \equiv M_0$.

Active entrepreneurs issue shares at a price of one (e.g., thousand dollars) per share. The amount they wish to raise is S , but the amount obtained will depend on investors' behavior which is influenced by their information (discussed in Section B below). The funds are raised through sealed submissions, which are made before total subscriptions are known. We assume that the market is competitive for the entrepreneurs.

B. Investors

A large number, N , of investors are searching for investment opportunities. Without loss of generality, we shall normalize investors' initial (investable) wealth to be one (e.g., thousand dollars) for all investors (by increasing or reducing the number of investors). Investors are also assumed to be well-diversified and they are only concerned about the expected return from the investment since all risks are firm-specific.¹ That is, investors' expected utility can be represented by $EU(R) = \bar{R}$, where $\bar{R}$ is the expected return or period ending wealth derived from the initial investment. If an investor commits to a project of quality q and the entrepreneur's perquisite consumption share is p , then $\bar{R} = (1 - p)\bar{X}(q)$.

Like many imperfect information models, we assume investors (as well as entrepreneurs) have rational expectations in equilibrium. The investors do not observe the quality of the projects or the entrepreneurs' perquisite consumption but they know the probability distribution of $\bar{R}$ in equilibrium.

Each investor has to decide, based on his knowledge about the $\bar{R}$ distribution,

¹ Investors may not be well-diversified in an imperfect market. Risk neutrality of investors will serve our purpose, but it may be a stronger assumption than we need. It is assumed that the entrepreneurs do not hold stock to signal the "quality" of investment. This problem has been analyzed by Leland and Pyle (1977). Rather, we focus on the question of how the entrepreneurial decisions are influenced by the acquisition of information by intermediaries.

whether or not to enter the venture capital market. If he avoids the venture capital market, he can invest in a risk-free asset which returns r per dollar (r is the opportunity cost of investment and it is assumed these risk-free investments are available in infinitely elastic supply). If he enters the market, he can learn about the attribute (p and q) of a project by dedicating resources to investigation. It is assumed that all investors perceive that, like investing in the risk-free asset, there is minimal (zero) effort for the first investigation. However, investors incur a utility-equivalent search cost, μ , for each project studied subsequently, due to additional effort required, cost of inconvenience, etc. Since there are a large number of projects in equilibrium, we assumed that investors will not revise their expectation on $\bar{R}$ despite their search activities.

Investors may have different investigatory skills and sources of information and hence different costs of search. As shown in the next section, the equilibrium distribution of (p, q) combinations will depend on the distribution of search costs across investors. For example, when $\mu = 0$ for all investors, only the optimal (p, q) combinations will be offered by entrepreneurs.

C. Optimal Investment Policy

Denote the equilibrium (p, q) distribution by $\phi(p, q)$ and let $\bar{R}_\phi$ be the expected return from investing in one randomly chosen project. We assume all investors will enter the market if and only if $\bar{R}_\phi > r$ (that is, if the venture capital market offers an expected return larger than the risk-free rate). The investor's search is assumed to proceed as follows. A project is randomly chosen and the (p, q) combination for that project is ascertained. The investor then either accepts the offer and invests in the project or rejects the offer and investigates another project, thereby incurring a search cost μ . Each (p, q) combination is assumed to be a random selection from $\phi(p, q)$. It can be shown that the optimal search policy for an investor with search cost μ is to reject all (p, q) offers that yield levels of utility below a single critical number, ϵ , and to accept any offer otherwise.² Formally, suppose an investor with search cost $\mu \geq 0$ selects a project (p', q') , then he will continue searching if and only if

$$\bar{R}' < \epsilon \quad (1)$$

where $\bar{R}' = (1 - p')\bar{X}(q')$ and ϵ is determined by

$$\int_{\bar{R} \geq \epsilon} (\bar{R} - \epsilon) d\phi(\bar{R}) = \mu. \quad (2)$$

where $\phi(\bar{R})$ is the distribution of $\bar{R}$ induced by the distribution $\phi(p, q)$, with $\bar{R}(p, q) = (1 - p)\bar{X}(q)$. Letting $P_\epsilon = P(\bar{R} \geq \epsilon)$ denote the probability of selecting a project with $\bar{R} \geq \epsilon$, Equation (2) can be rewritten as:

$$\left[\frac{1}{P_\epsilon} \int_{\bar{R} \geq \epsilon} \bar{R} d\phi(\bar{R}) \right] - \epsilon = \frac{\mu}{P_\epsilon}. \quad (3)$$

The first term in the left-hand side of (3) is the expected utility from following

² See McCall (1970) for a derivation of the optimal search policy in the context of labor market.

the optimal policy to search for projects until the investor finds a project with $\bar{R} > \epsilon$. The left-hand side of (3) therefore represents the expected benefit of searching optimally after the investor has selected a project that yields ϵ , and the right-hand side is the expected search cost of finding a project that yields ϵ or better, since $1/P$ is the expected number of search before the investor will find a project with return $\bar{R} \geq \epsilon$. The optimal search policy will set ϵ such that the expected benefit is equal to the expected cost. Note from (2) that ϵ is determined by $\phi(\bar{R})$ and μ ; and given ϕ , ϵ is decreasing in μ . The higher the search cost, the lower will be the critical level of utility. When μ is sufficiently large such that $\epsilon = \min\{\bar{R} \text{ offered in equilibrium}\}$, the investor will find it optimal to invest randomly. With $\mu = 0$, it is assumed that the investors will investigate all firms and invest in the projects that yield the highest expected return.

D. Definitions

Let $p(I)$ and $q(I)$ be the solution to the following problem, given $I(I \leq S)$:

$$\max_{p,q} (1 - p) \bar{X}(q)$$

subject to

$$M[pI \bar{X}(q), e(q)] \geq M_0. \quad (4)$$

$p(I)$ and $q(I)$ are the perquisite consumption share and project quality an entrepreneur selects in order to maximize the investors' utility subject to the condition that the entrepreneur is at least as well off as he would be by not undertaking any project. The solutions $p(I)$, $q(I)$ will satisfy the first-order conditions:

$$M_1 I \bar{X}'(q) + M_2 e'(q) = 0, \quad (5)$$

$$M[pI \bar{X}(q), e(q)] = M_0. \quad (6)$$

The second-order conditions for a maximum are assumed to be satisfied. Define $V(I)$ as the maximum in problem (4) and

$$p^* \equiv p(S) \quad (7)$$

$$q^* \equiv q(S) \quad (8)$$

$$V^* \equiv V(S). \quad (9)$$

In a world with perfect (costless) information and a competitive market for entrepreneurs, only (p^*, q^*) projects will be offered by the entrepreneurs.

II. Search Costs and Equilibrium Distribution of Perquisite Consumption and Qualities of Projects

In deriving the equilibrium distribution (p, q) combinations when there is costly information, two cases³ are considered:

- i) $\mu_1 = 0$ for αN investors and $\mu_2 > 0$ for $(1 - \alpha)N$ investors; and

³ The results generalize for any arbitrarily chosen number of investor classes.

- ii) $\mu_2 > \mu_1 > 0$ with α and $(1 - \alpha)$ proportion, for group 1 and group 2, respectively.

A. Equilibrium

Equilibrium is characterized by a set of perquisite/quality combinations

$$\{(p_1, q_1); (p_2, q_2); \dots; (p_K, q_K)\}$$

selected by the entrepreneurs. Associated with any equilibrium will be the endogenously determined number of entrepreneurs $(M_1, M_2, \dots, M_K)$ selecting these combinations, and the endogenous searching behavior of the investors. The following conditions must be satisfied in equilibrium.

(E1) *Optimal Investment Decisions*: An investor with positive search cost μ_i will follow the optimal search policy described in Section I C if $\epsilon_i > \min_k \{\bar{R}_k\}$, where $\bar{R}_k = (1 - p_k)\bar{X}(q_k)$ and ϵ_i is determined in Equation (2) when $\mu = \mu_i$ and $\phi(\bar{R})$ is the equilibrium distribution of returns induced from the equilibrium distribution of perquisite/quality $\phi(p, q)$ which is characterized completely by the sets $\{(p_k, q_k)\}$ and $\{M_k\}$, $k = 1, 2, \dots, K$. We shall call these investors “searchers.” When $\epsilon_i = \min\{\bar{R}_k\}$, the investor i will invest in a project randomly chosen and we refer to such investors as “random investors.”⁴

(E2) *Utility Maximization for Entrepreneurs*: The (p_k, p_k) combination chosen by entrepreneur k yields at least as much utility as any other (p, q) combination he could offer, given

- (a) imperfect information in the market—investors do not observe the (p, q) without sustaining search costs, μ ,
- (b) Nash behavior vis-à-vis other entrepreneurs, and
- (c) Stackelberg behavior vis-à-vis investors.

Conditions (b) and (c) are rules that an entrepreneur follows when considering a possible change in (p, q) in order to increase his utility. Condition (b) says that the entrepreneur takes the behavior of other entrepreneurs as given independently of his decisions. Condition (c) says that the entrepreneur takes the reactions of the investors to his decisions into account to see whether it is worthwhile to deviate. Condition (a) indicates that investors will not know about any changes of a particular entrepreneur unless they happen to investigate his activities.⁵ A set of (p, q) combinations, $\{(p_k, q_k)\}$, together with the associated set $\{M_k\}$ will constitute an equilibrium only if there is no economic incentive for any entrepreneur to deviate from his chosen (p, q) combinations.

(E3) *All Entrepreneurs Enjoy the Same Utility*, M_0 : This is a result of free entry to and exit from the market.

(E4) *The Supply of Investment Funds to the Projects is Determined by the Law*

⁴ Note that in equilibrium there must be some random investors who find it optimal to invest in the projects with the lowest return. Otherwise, the sponsors of these projects will not obtain any funds and be worse off than not undertaking the project, thus violating the conditions for equilibrium.

⁵ If an entrepreneur offers a new (p, q) combination, the investors who happen to investigate his activities are assumed to use an unchanged prior, $\phi(\bar{R})$.

of Large Numbers: Since each project investigated is a random selection from $\phi(p, q)$, the number of investors that a project will attract is random. Since in equilibrium there will be a large number of investors and entrepreneurs the law of large numbers is assumed to hold. Hence the realized supply of funds equals the expected supply of funds to the projects, and projects yielding the same expected return will obtain the same amount of investment funds.

B. Equilibrium When Some Investors Face Zero Search Cost: $\mu_1 = 0$

PROPOSITION 3.1. *The unique combination (p^*, q^*) will be offered in equilibrium if and only if $\mu_2 < V^* - V[(1 - \alpha)S]$.*

Proof: See Appendix A.

The equilibrium (p^*, q^*) is the same as in the competitive case with costless information. There is an externality effect of the existence of zero-cost μ_1 investors on the welfare of the high cost μ_2 consumers. Since only the best (p, q) combinations are offered in equilibrium, everyone can invest randomly and realize V^* .

Let $Y = V^* - V[(1 - \alpha)S]$ and $V =$ utility from a project with the lowest equilibrium expected return. We have the following two (p, q) combinations equilibria:

PROPOSITION 3.2. *The combinations (p^*, q^*) and (p', q') will be offered in equilibrium if and only if $\mu_2 \geq Y$, where $p' = p(I')$, $q' = q(I')$, and I' is the investment in projects offering (p', q') in equilibrium. Let M be the total number of projects, M^* be the number of (p^*, q^*) projects, and $\lambda = M^*/M$. The equilibrium distribution of (p, q) combinations will satisfy the following conditions:*

- (i) $S = (\alpha N/M^*) + [(1 - \alpha)N/M]$;
- (ii) $I' = (1 - \alpha)N/M$ ((i) and (ii) are market-clearing conditions); and
- (iii) (a) $V^* - (\mu_2/\lambda) = V$ when $Y \leq \mu_2 \leq Z$; or
 (b) $V^* - (\mu_2/\lambda) < V = r$ when $\mu_2 > Z$

where Z is a positive number determined in Appendix B, and $V = (1 - p')\bar{X}(q')$.

Proof: See Appendix A.

Note that in equilibrium the $\mu_1 = 0$ investors will investigate all projects and choose the (p^*, q^*) combination yielding utility V^* . Condition (iii) implies that there is no incremental benefit from searching for a V^* project once a μ_2 investor has discovered a project that yields a return V . ($V^* - (\mu_2/\lambda)$ is the expected utility of searching sequentially for a V^* project.) Therefore, the μ_2 investors are random investors in equilibrium and their expected utility is $\bar{V} = \lambda V^* + (1 - \lambda)V$ (see Appendix B). When $\mu_2 < Z$, V exceeds r , the utility offered by the risk-free asset. But when $\mu_2 > Z$, V equals r .

Some comparative statics results: In Appendix B it is shown that λ and V will increase as α increases or μ_2 decreases, other things equal. Thus, more informed investors or lower search costs improves the allocation of resources.

C. Equilibrium When All Investors Have Positive Search Costs

PROPOSITION 3.3. *When all investors have positive search costs none will enter the venture capital market, rather, they will all choose the risk-free asset.*

Significant change occurs in equilibrium results when μ_1 is positive. When all investors have positive search costs, the entrepreneurs will try to extract all the surplus from the uninformed and there will be a degenerate distribution of (p, q) combinations that yield a return r , the same return offered by the risk-free asset. Investors will therefore avoid the venture capital market. Note that the existence of zero-cost investors insures that (p^*, q^*) combinations will be offered by entrepreneurs because the zero-cost investors will always discover the “best buy” in the market. Conversely, their existence will discourage the (p^*, q^*) entrepreneurs from offering any other combination because in doing so they will lose all the zero-cost investors. With $\mu_1 > 0$, the μ_1 investors will search if and only if the expected benefit exceeds the expected cost. Suppose a proposed equilibrium $\{(p_1, q_1), (p_2, q_2)\}$ with $(1 - p_1)\bar{X}(q_1) > (1 - p_2)\bar{X}(q_2)$ and the distribution of (p, q) are such that μ_1 investors will search for (p_1, q_1) projects in equilibrium. It is easy to check that $\{(p_1, q_1), (p_2, q_2)\}$ cannot be an equilibrium since an entrepreneur offering (p_1, q_1) can increase his perquisite consumption by a sufficiently small percentage without losing the μ_1 investors who happen to investigate the project (randomly). Therefore, when $\mu_1 > 0$, multiple (p, q) equilibria do not exist with both searchers and random investors. Entrepreneurs will extract all the surplus and with rational expectations the investors will avoid the market because they know only the “lemons” are offered. This is analogous to Akerlof’s “lemons problem” with prohibitive information cost.

D. Policy Implications

The equilibrium distribution of (p, q) combinations depends critically on the distribution of search costs of the investors. When all investors have positive search costs, only “lemons” are offered and investors will not enter the market. When some investors have zero search costs, there is an improved allocation of resources in terms of the entrepreneurs’ efforts and scale of projects. In particular, when μ_2 is sufficiently small, the welfare of all investors will be the same as in the case of costless information. An interesting question is: *suppose in the initial situation all investors have positive search costs, are there any policies or institutions that can lower the search cost of some agents to zero; thereby leading the economy to a higher welfare state, net of the costs of any such policies?* Section III examines the remedial role of financial intermediation for inferior resource allocations resulting when all investors face positive search costs.

III. A Model of Financial Intermediation

A. An Integrated Theory of Financial Intermediation

Extant explanations for the emergence of financial intermediaries can be classified in terms of the roles of the intermediaries, their functions, or contri-

butions as follows:

- (i) reduction of transaction costs (which may include costs of collection and analysis of information), e.g., Benston (1972), Benston and Smith (1977), Flannery (1974, 1978), Gurley and Shaw (1960);
- (ii) portfolio optimization and asset transformation, e.g., Klein (1971), Pyle (1971, 1972), Deshmukh et al. (1983a, b), Diamond and Dybvig (1983);
- (iii) information producer/signaling agent, e.g., Leland and Pyle (1977), Draper and Hoag (1978), Campbell and Kracaw (1980); and
- (iv) monitoring agent, e.g., Diamond (1980, 1982).

These are all plausible explanations for intermediation (see Baltensperger (1980)). However, except for the transaction cost approach none have been able to attach explicitly a positive role to intermediaries. For instance, we note earlier that there is no net contribution from the production of information by the Campbell-Kracaw's intermediaries, instead there is a dead weight loss of information costs.

The present model analyzes the role of intermediaries in a world of costly information, recognizing that intermediaries conserve societal search costs. Further, the model recognizes that intermediaries increase the investors' welfare by inducing a Pareto-preferred allocation of resources. The model integrates the transaction cost approach and the information-based approach to financial intermediation, focusing on positive contribution of intermediaries. This approach highlights the well-known result that information may have no social value if it fails to alter production decisions (see Hirshleifer (1971) and Hakanson et al. (1982)).

B. Characterization of Intermediation Equilibrium

Suppose our scenario is a world where all N investors have positive search costs. We shall first consider the simple case where all investors have the same search cost μ (as in Grossman and Stiglitz (1980)). In Section III *D* we consider the case with differential search costs.

From the analysis in Section II, we know that without information acquisition the investors will not enter the venture capital market. Now suppose some investors decide to form financial intermediaries that search for good projects and invest for the clients for a fee, d , per individual.^{6,7}

- (i) We assume that each intermediary has the same constant search cost μ of the investors. Once an intermediary has identified a V^* firm, it can accommodate S investors. Therefore, an intermediary has the comparative advantage in information production due to the reusability of infor-

⁶ In order to avoid discounting, the service fee d is assumed to be assessed at time 1, the time when the total return of investment $\tilde{X}(q)$ is realized.

⁷ In order to focus on the positive contribution of intermediaries in this setting, we do not consider the moral-hazard problem of the reliability of information production. In our model, the reliability problem can be resolved if some kind of monitoring or binding contract on managers is feasible [see Diamond (1982)]. In this case, the monitoring costs have to be added to the search costs to obtain the total operating costs of the intermediaries.

mation.⁸ Note, however, that such advantage in information production is not necessary for intermediation to be valuable to the economy. As explained below, as long as the intermediaries' information costs are paid by clients, the intermediaries play the role of zero-cost, informed agents who induce preferred allocations. In general, the intermediaries' average search cost per project may differ from μ , depending on the cost structure of the intermediaries.

- (ii) The intermediaries divide up the total number of projects randomly among themselves for investigation, that is, every intermediary has an equal share of the "good" and "bad" projects and there is no duplication of effort. (This is assumed for simplicity only. The essential point is that all projects are investigated (randomly) by intermediaries; to the extent that there is duplication of effort, the opportunity to invest in the project (if desired) is shared equally among the intermediaries involved. The cost of duplication of effort will, of course, be reflected in the average search cost function.)
- (iii) The intermediaries also secure an equal share of the investors who place their funds in intermediaries.
- (iv) The market for intermediaries is competitive.

Note that since all intermediaries are identical, the investors will choose the intermediary with the lowest fee. Thus, in equilibrium there will be a uniform fee, d , among all intermediaries.

We characterize a *competitive intermediation equilibrium* by the following:

- (i) the (p, q) combinations and the number of each of them selected by the entrepreneurs, $\phi(p, q)$;
- (ii) the proportion of investors, γ , who use the intermediaries, which represents the fraction of "pseudo" zero-cost investors (since their costs are paid by their clients) in equilibrium which induces $\phi(p, q)$;
- (iii) the fee charged by the intermediaries, d ; and
- (iv) the number of intermediaries, h .

The existence of intermediaries as "pseudo" zero-cost investors will induce a distribution of (p, q) combinations in equilibrium by either of the following forms: (i) (p^*, q^*) ; or (ii) $\{(p^*, q^*); (p', q')\}$, as described in Propositions 3.1 and 3.2. Whether form (i) or (ii) will obtain depends on the parameters in the model: $U(\cdot)$; $M(\cdot, \cdot)$; $\bar{X}(\cdot)$; $e(\cdot)$; $\mu_2 (= \mu)$; $\alpha (= \gamma)$; and N , and on the equilibrium conditions.

The following conditions must be satisfied in the equilibrium:

- (IE1) All entrepreneurs maximize their utilities with the behavior described in (E2) (Section II A).
- (IE2) All entrepreneurs have the same utility $M_0(E3)$.
- (IE3) The investors make optimal decisions; an investor with search cost μ (>0) will either (i) adopt the random investment strategy, or (ii) invest

⁸ I thank Stuart Greenbaum and George Kanatas for making this point to me.

in the intermediaries and receive an expected return $V^* - d$.⁹ Since the investors have identical search cost, μ , and face the same opportunities in equilibrium, the distribution of (p, q) combination must be such that investors will be indifferent as to which policy they choose, that is, $V^* - d = \bar{V}$, where $\bar{V}$ denotes the expected utility from the random investment policy.

(IE4) The intermediaries maximize their profits by charging the optimal service fee, given the fees charged by other intermediaries.

(IE5) The intermediaries earn zero profit.

C. Determination of Competitive Intermediation Equilibrium When All Investors Have the Same Search Cost μ

Suppose $\{\phi(p, q); \gamma; d; h\}$ is a competitive intermediation equilibrium (hereafter referred to as a CI equilibrium) in which a proportion, γ , of investors with search cost, μ , patronize the intermediaries. The induced equilibrium distribution of (p, q) combinations, $\phi(p, q)$, is determined as if there were γN zero-cost investors and $(1 - \gamma)N$ investors with search cost μ (using the notations in Section II, $\mu_1 = 0$, $\mu_2 = \mu$, and $\alpha = \gamma$). Let

$\lambda(\gamma)$ = proportion of (p^*, q^*) projects in the CI equilibrium;

$M(\gamma)$ = total number of projects in the CI equilibrium;

$V(\gamma) = (1 - p')\bar{X}(q') =$ expected return on inferior projects in the CI equilibrium; and

$I(\gamma)$ = investment in inferior projects in the CI equilibrium,

then

$$V^* - d = \bar{V} = \lambda(\gamma)V^* + (1 - \lambda(\gamma))V(\gamma) \quad (10)$$

Since the intermediaries are assumed to have constant search cost, the number of intermediaries in equilibrium is indeterminate. Suppose the number is h^* , then the number of projects investigated by each intermediary is $n^* = M(\gamma)/h^*$.

The total revenue of intermediaries is γdN . Let $TR(\gamma)$ be the revenue of each intermediary, then

$$TR(\gamma) = \frac{\gamma dN}{h^*} = \frac{\gamma d n^* N}{M(\gamma)} \quad (11)$$

Define $F(\gamma) \equiv TR(\gamma)/n^*$. $F(\gamma)$ measures the willingness to pay (per project) for intermediation service of investors. The *zero-profit condition* yields:

$$F(\gamma) = \frac{\gamma d N}{M(\gamma)} = \mu \quad (12)$$

Equation (12) implies d is a function of γ and we can rewrite $F(\gamma)$ as:

$$F(\gamma) = \frac{\gamma d(\gamma) N}{M(\gamma)} \quad (13)$$

⁹ From the discussion following Proposition 3.2, when there are only two classes of investors ($\mu_1 = 0$, $\mu_2 > 0$), in equilibrium the high search-cost investors will not search for V^* projects at their own cost.

From Proposition 3.1 we note that the unique combination (p^*, q^*) will be offered if and only if $V^* - V[(1 - \gamma)S] > \mu$. This condition is equivalent to $\gamma > 1 - [V^{-1}(V^* - \mu)/S]$. Define

$$\gamma^* \equiv 1 - [V^{-1}(V^* - \mu)/S]. \quad (14)$$

Then when $\gamma > \gamma^*$ the only combinations offered are (p^*, q^*) . An individual investor would then be foolish to engage an intermediary since he could do as well by investing randomly and thereby saving the intermediary's fee. Therefore when $\gamma > \gamma^*$, $d(\gamma) = 0$ and $F(\gamma) = 0$, and intermediation is not economically viable since the information cost of intermediaries is positive.

When $\gamma = 0$, the entrepreneurs are induced to offer inferior projects with a return r and investors will avoid the venture capital market. In this case, with rational expectations, no investor is willing to use the intermediaries since he knows the intermediaries cannot possibly find projects that yield greater than r . Therefore $F(0) = 0$.

Let $F^* \equiv \max\{F(\gamma), 0 \leq \gamma \leq 1\}$ and γ_m be the maximizer of F in the interval $(0, 1)$. That is, $F^* = F(\gamma_m)$. Appendix C shows that γ_m is unique and $F(\gamma)$ is increasing in $(0, \gamma_m)$ and decreasing in (γ_m, γ^*) (as discussed above $F(\gamma) = 0$ for $\gamma \geq \gamma^*$). $F(\gamma)$ also depends on μ as follows.

(i) When $\mu > (V^* - r)/2$,

$$\gamma_m = (S - I_r)/(S + I_r) \quad (15)$$

and

$$F^* = (V^* - r)(S - I_r)/4, \quad (16)$$

where I_r is the level of investment such that $V(I_r) = r$ as defined in Appendix B.

(ii) When $\mu \leq (V^* - r)/2$,

$$\gamma_m = \left[1 + \frac{V^* - r}{\mu} \frac{I_r}{S - I_r} \right]^{-1} \quad (15a)$$

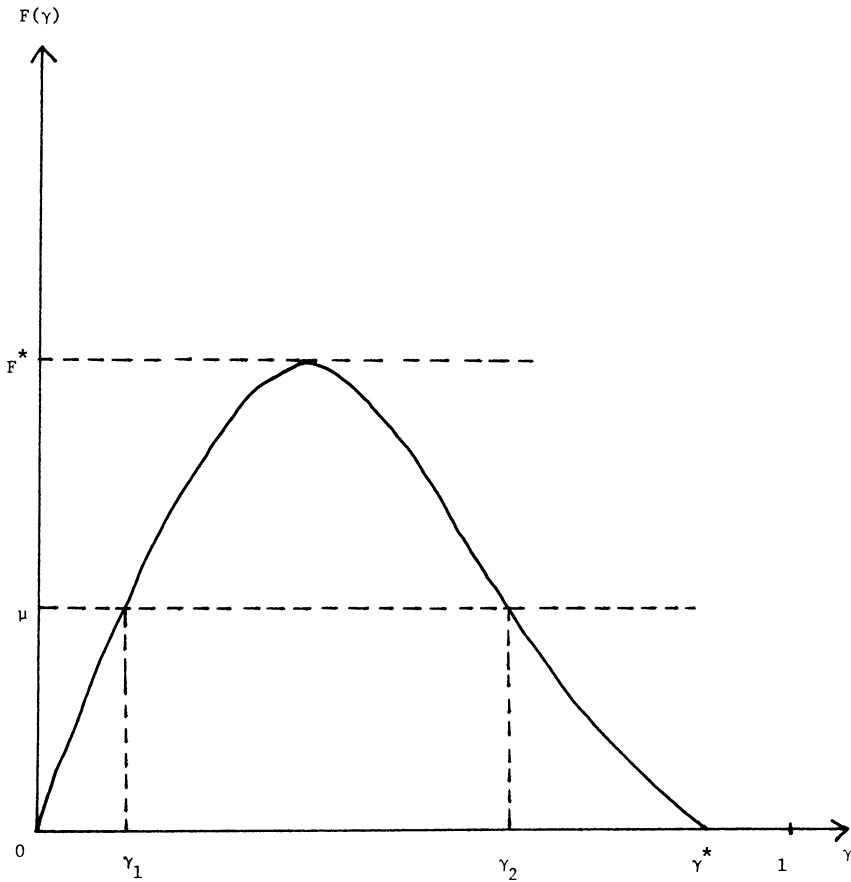
and

$$F^* = (S - I_r)[1 - (\mu/(V^* - r))]\mu \quad (16a)$$

A typical form of $F(\gamma)$, for given μ , is shown in Figure 1 and F^* as a function of μ is shown in Figure 2.

PROPOSITION 4.1. *A CI equilibrium does not exist in which $\{1 - [V^{-1}(V^* - \mu)/S]\}N$ or more investors choose to invest in the intermediaries.*

This result is analogous to Grossman and Stiglitz' (1980) observation that there cannot be an informationally efficient market since if the market is efficient those who expend resources to obtain information will receive no compensation. The analyses in both models reflect a fundamental conflict between the induced efficiency of the market (from information acquisition) and the incentive to acquire information. The difference is that in Grossman and Stiglitz' model, information acquisition by some individuals makes the price system more inform-



NOTE: $\gamma^* = 1 - [V^{-1}(V^* - \mu)/S]$

Figure 1. Determination of Competitive Intermediation Equilibria (for given μ)

ative (informational efficiency) whereas in our model the search activities of the intermediaries induce a higher degree of allocational efficiency.

PROPOSITION 4.2. Define $\mu^* \equiv (V^* - r)[1 - (1/(S - I_r))]$ such that $\mu^* = F^*(\mu^*)$.

- (a) When $\mu \leq \mu^*$, there exists a CI equilibrium (which may not be unique).
- (b) When $\mu > \mu^*$, a CI equilibrium does not exist.

Proposition 4.2 states that when the total willingness to pay for intermediation services is sufficient to cover the intermediaries' costs a CI equilibrium will exist; otherwise, a CI equilibrium will not exist. Note that $V^* - r$ and $S - I_r$ measure the improvements with intermediation and since μ^* is increasing in them, the proposition indicates that intermediation is more likely to be viable when their contribution is large.

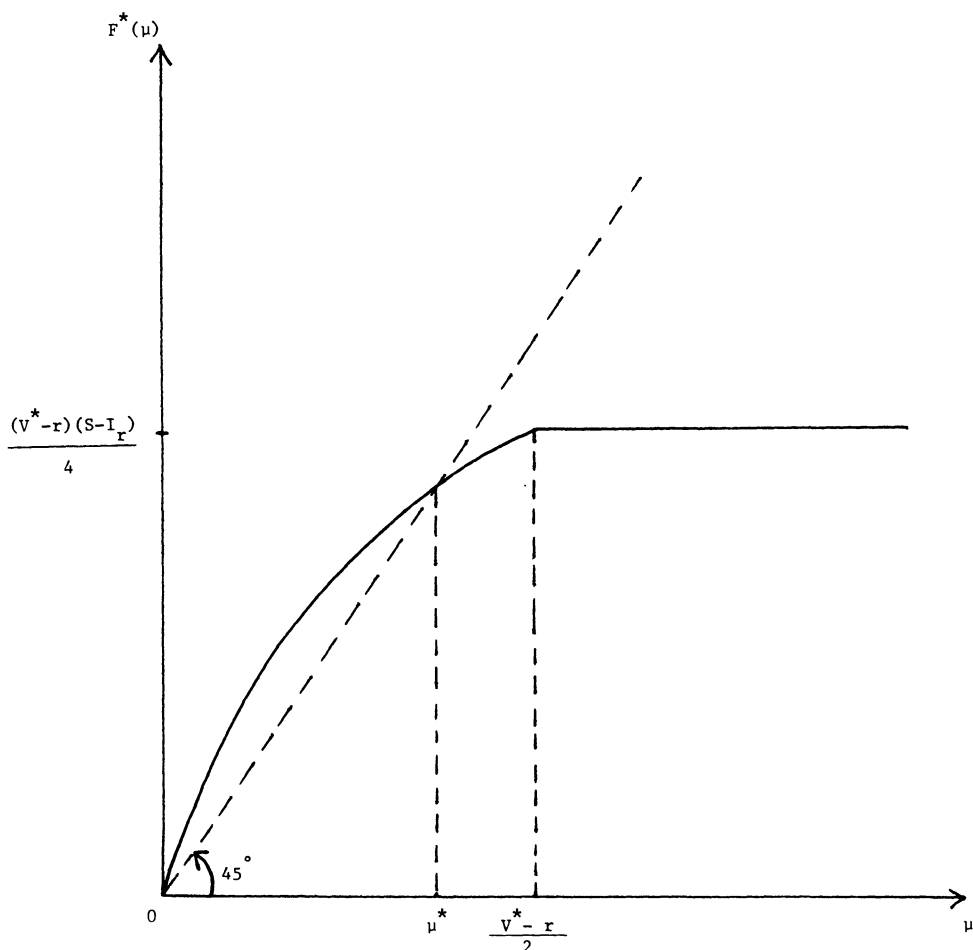


Figure 2. The Maximum Willingness to Pay Function F^*

PROPOSITION 4.3. *When there are multiple CI equilibria, the larger the proportion of investment funds held by the intermediaries (γ), the higher is the level of utility attained by all investors—equivalently, an equilibrium with a high γ will dominate that with a small one using a Pareto criterion.*

Proof: In Appendix B.

Considering the intermediaries as informed investors in the market, Proposition 4.3 indicates that the more informed is the market, the better off are the investors. This makes sense because a more informed population will induce a better resource allocation by the entrepreneurs. The economic reasons behind this are two-fold. When more “investors” are informed, (i) the number of projects undertaken is smaller because I is closer to S , therefore requiring smaller expenditure on information costs; (ii) the average return of the projects is higher because there is a higher proportion of V^* projects and a higher return on the

other projects offered in equilibrium. As a result, the welfare of the investors increases.

D. Competitive Intermediation Equilibria with Differential Search Costs

In this section, we consider a more general environment where there are differential search costs among investors. Suppose there are two types of investors, βN of them have a search cost $\mu > 0$ (as in previous sections) and $(1 - \beta)N$ of them have a search cost $\delta > \mu$.¹⁰

In the following, we assume the intermediaries have the same information cost μ as the low cost investors. From Proposition 4.1, a CI equilibrium exists only if some, but not too many, investors patronize the intermediaries. With two types of investors, a CI equilibrium exists in which, given the equilibrium (p, q) distribution and intermediary fees, either (i) the investors with low search cost are indifferent between patronizing the intermediaries or investing randomly (which implies that investors with high search costs will find it optimal to put their funds in intermediaries); or (ii) the high search-cost investors are indifferent between patronizing the intermediaries or investing randomly (which implies that the low search-cost investors will find it optimal to search for the high return projects on their own).

In case (i) all of the high search-cost investors and an (equilibrium) proportion of low search-cost investors will put their funds in the intermediaries and the other low search-cost investors will invest randomly in equilibrium. In case (ii) only an (equilibrium) proportion of high search-cost investors patronize the intermediaries; the other high search-cost investors will invest randomly, while the low search-cost investors search sequentially for V^* projects at their own cost.

Since our purpose is to illustrate that CI equilibria exist with differential search costs among investors, we shall limit our discussion to the equilibria in case (i).¹¹

Let w = proportion of low search-cost investors who patronize the intermediaries in equilibrium and γ = percentage of institutional holdings, then for equilibria of type (i),

$$\gamma = w\beta + (1 - \beta)$$

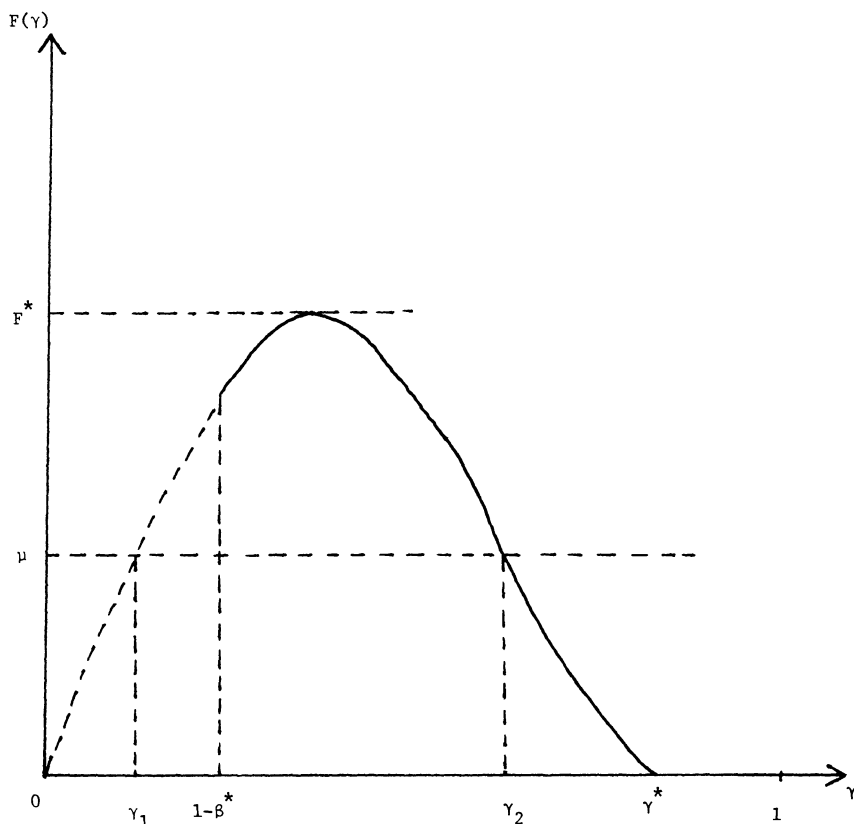
or

$$w = \frac{\gamma - (1 - \beta)}{\beta}.$$

Hence w can be determined when γ is known. But the value of γ can be determined as in Section III C using (10), (11), and (12). The only condition that has to be satisfied is $w \geq 0$, which implies $\gamma > 1 - \beta$ (i.e., $\beta > 1 - \gamma$), for any

¹⁰ The analysis can be extended to the case of many types of investors.

¹¹ For a type (ii) CI equilibrium, essentially three classes of investors exist in the market: the intermediaries who are pseudo zero-cost investors, the low search-cost investors, and the high search-cost investors who invest randomly. It can be shown that there will be an equilibrium with either two or three (p, q) combinations.



NOTE: $\gamma^* = 1 - [V^{-1}(V^* - \mu)/S]$

Figure 3. Determination of Competitive Intermediation Equilibria When There are Differential Search Costs (for given μ)

equilibrium value γ determined in Section III C. This is illustrated in Figure 3. In equilibria of type (i), since all the high search-cost investors will patronize the intermediaries, only the solid-lined portion of the curve $F(\gamma)$ domain: $\gamma > 1 - \beta$ is applicable. For a given value of $\beta = \beta^*$, the CI equilibrium with a low institutional holding of γ_1 does not exist, but the one with a high institutional holding of γ_2 does. The underlying reason is that in a type (i) equilibrium the high search-cost investors will patronize the intermediaries; a large number of these investors may increase the institutional holdings in a CI equilibrium, which induce Pareto-preferred allocations. However, welfare will not necessarily increase with more high cost investors. When there are too many high search-cost investors such that $1 - \beta > \gamma_2$, a type (i) equilibrium does not exist. The reason is that given the intermediaries' cost, an institutional holding higher than γ_2 cannot generate sufficient service fees to compensate the intermediary.

IV. Conclusion

We have shown that, in a venture capital market where all investors have positive information costs, the entrepreneurs are induced to offer inferior projects and the investors will avoid the market and put their funds in other low return investments resulting in an undesirable allocation of resources. This is because without informed investors the entrepreneurs will find it in their interests to offer less desirable projects leading to the degeneration of the projects undertaken. In a market containing some informed investors, the allocation will be Pareto-preferred, but the improvement is limited by the proportion of informed investors and the magnitude of information costs of the other investors. It is shown that in equilibrium the higher the search costs of the uninformed investors, the lower will be the average return of the projects offered by entrepreneurs.

Beginning with an initial situation with imperfect information, we explain how intermediaries, as informed agents, will increase the welfare of investors by inducing the entrepreneurs to offer high return projects. We focus on the existence of competitive intermediation (CI) equilibria and the ranking of such equilibria with the Pareto criterion. It is shown that there cannot be a CI equilibrium with very high institutional holdings. Further, we show that there are cases where multiple CI equilibria may exist, but the one with the highest institutional holdings will dominate the others in a Pareto sense.

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Appendix A

Proofs of Propositions

A. We shall first establish that the (p, q) combinations selected by the entrepreneurs receiving an investment of I' at equilibrium must be $p' = p(I')$ and $q' = q(I')$, where the functions $p(I)$ and $q(I)$ are the solutions to the maximization problem (4).

The proof is as follows. In equilibrium all entrepreneurs have the same utility M_0 , and they will either receive the investment funds (say I') from (i) the uninformed investors only or (ii) from some informed (zero-cost) investors and some uninformed investors (who invest randomly in one project). Suppose (i) is the case, and suppose that there are entrepreneurs who offer $(p, q) \neq (p', q')$ to the uninformed investors only. An entrepreneur offering (p, q) will find it in his interest to switch to some other combination(s) $(\bar{p}, \bar{q})$ such that $(1 - \bar{p})\bar{X}(\bar{q}) \geq (1 - p)\bar{X}(q)$, so that he would not lose any uninformed investors (who are

assumed to be satisfied with (p, q) in equilibrium) and, simultaneously, increase his welfare from M_0 to $M[I'\bar{p}\bar{X}(\bar{q}), e(\bar{q})]$. Thus (p, q) cannot be offered in equilibrium.

Now suppose (ii) is the case, and suppose there are entrepreneurs who offer $(p, q) \neq (p', q')$ to the informed and some (lucky) uninformed investors and receive total investments of I' . It is obvious that an entrepreneur will find it in his interest to offer some combination(s), $(p^* + \epsilon, q^*)$, which will attract all the informed investors he needs (S is the maximum) to increase his welfare from M_0 to $M[S(p^* + \epsilon)\bar{X}(q^*), e(q^*)]$. Thus the proof is complete.

B. Proof of Proposition 3.1: (i) When $\mu_1 = 0$, from (A) the entrepreneurs offering projects to the zero-cost investors must offer (p^*, q^*) in equilibrium. When $\mu_2 < V^* - V[(1 - \alpha)S]$, it is shown in Appendix B that $\{(p^*, q^*)\}$ is the only equilibrium that exists. (ii) Suppose all entrepreneurs offer (p^*, q^*) . We shall verify that it is indeed an equilibrium. $\{(p^*, q^*)\}$ will be an equilibrium if there is no incentive for any entrepreneur to deviate from (p^*, q^*) . The entrepreneurs know that if they deviate from (p^*, q^*) they will lose all the zero-cost investors and the remaining number of μ_2 investors (if they desire not to search for other projects) is $(1 - \alpha)S$. Therefore, when $\mu_2 < V^* - V[(1 - \alpha)S]$, no entrepreneurs will deviate because they cannot keep the μ_2 investors without lowering their utility below M_0 .

C. Proof of Proposition 3.2: (i) When $\mu_1 = 0$ and $\mu_2 > 0$, from (A) if equilibrium exists, it must be of the form (p^*, q^*) , (p', q') . Appendix B shows that the equilibrium as characterized by conditions (i), (ii), and (iii) in Proposition 3.2 for different values of μ_2 is unique when certain conditions are satisfied.

(ii) It is easy to verify that $\{(p^*, q^*), (p', q')\}$ is indeed an equilibrium. $\{(p^*, q^*), (p', q')\}$ will be an equilibrium if there is no incentive for any entrepreneur to deviate from his prevailing perquisite/quality combination. First note that the (p^*, q^*) entrepreneurs have no incentive to deviate, because if they do they will lose all the zero-cost investors and the remaining number of μ_2 investors (if they can keep them after the deviation) is I' . But they have to offer at least $V(I') = (1 - p')\bar{X}(q')$ to keep the μ_2 investors, which will not make them better off than the current situation. Also, the (p', q') entrepreneurs have no incentive to change their perquisite quality combinations. They will be better off by doing so if they can still keep the μ_2 investors and at the same time increase their utility, which is impossible from conditions (iii)(a) and (iii)(b). If they increase V , they can keep the μ_2 investors, but at a loss; if they decrease V , they will lose all the μ_2 investors.

D. Proof of Proposition 3.3: In text.

E. Proofs of Propositions 4.1 and 4.2: In text.

F. Proof of Proposition 4.3: In Appendix B.

Appendix B

Derivation of Equilibria and Comparative Statics Analysis

Our purpose in this appendix is to derive the equilibrium distribution of (p, q) combinations selected by the entrepreneurs when there are some investors with

zero-search cost (αN of them) and some with positive cost μ ($(1 - \alpha)N$ of them), and then relate the equilibrium to the exogenous parameters.

When there are some zero-cost investors (who are informed), the equilibrium that will obtain is either (i) $\{(p^*, q^*)\}$ or (ii) $\{(p^*, q^*), (p', q')\}$, as described in Propositions 3.2 and 3.3. We have shown in the proofs that the necessary condition for the (p^*, q^*) equilibrium is $\mu < V^* - V[(1 - \alpha)S]$. Now we shall consider equilibrium (ii) and show that the distribution of $\{(p^*, q^*), (p', q')\}$ depends on μ , and as μ converges to $V^* - V[(1 - \alpha)S]$, equilibrium (ii) will converge to equilibrium (i), therefore the sufficiency of $\mu < V^* - V[(1 - \alpha)S]$ is established.

The following conditions must be satisfied in equilibrium (ii):

$$S = \frac{\alpha N}{M^*} + \frac{(1 - \alpha)N}{M} \quad (\text{B1})$$

$$I' = \frac{(1 - \alpha)N}{M} \quad (\text{B2})$$

$$\lambda[V^* - V(I')] = \mu. \quad (\text{B3})$$

Substituting (B2) in (B1) and using $\lambda = M^*/M$, we have,

$$I' = \frac{S}{\left(\frac{\alpha}{1 - \alpha}\right) \frac{1}{\lambda} + 1}. \quad (\text{B4})$$

Differentiating (B4) with respect to λ ,

$$\frac{dI'}{d\lambda} = S \left(\frac{\alpha}{1 - \alpha} \right) \frac{I'^2}{\lambda^2} > 0 \quad (\text{B5})$$

and

$$\frac{d^2 I'}{d\lambda^2} = \frac{2I'^2 S}{\lambda^4} \left(\frac{\alpha}{1 - \alpha} \right) \left[SI' \left(\frac{\alpha}{1 - \alpha} \right) - \lambda \right]. \quad (\text{B6})$$

So $d^2 I'/d\lambda^2 > 0$ if and only if $\lambda < (S^2 - 1)\alpha/(1 - \alpha)$.

V is a function of I' , and we write

$$V(\lambda) = V \left[\frac{S}{\left(\frac{\alpha}{1 - \alpha}\right) \frac{1}{\lambda} + 1} \right]. \quad (\text{B7})$$

Since V is increasing in I' , and I' is increasing in λ , $V(\lambda)$ is increasing in λ , that is, $V'(\lambda) > 0$. The second derivative of $V(\lambda)$, however, will depend on the shapes of $V(I')$ and $I'(\lambda)$.

Define I_r as the level of investment that yields the lowest return r , that is,

$$V(I_r) = r. \quad (\text{B8})$$

Define λ^* such that $I' = I_r$ in (B4), that is,

$$\lambda^* = \frac{\alpha}{1 - \alpha} \frac{I_r}{S - I_r}. \quad (\text{B9})$$

Equation (B3) also relates V to λ . Let $Q(\lambda, V, \mu) = \lambda[V^* - V] - \mu$, then (B3) can be rewritten as

$$Q(\lambda, V, \mu) = 0. \quad (\text{B10})$$

Define $\underline{\lambda}$ as the value of λ when $V = r$, that is,

$$\underline{\lambda} = \mu / (V^* - r). \quad (\text{B11})$$

The derivatives of the curve $Q = 0$ are:

$$\left. \frac{dV}{d\lambda} \right|_{Q=0} = \frac{V^* - V}{\lambda} > 0 \quad \text{and} \quad \left. \frac{d^2V}{d\lambda^2} \right|_{Q=0} = -\frac{2(V^* - V)}{\lambda} < 0.$$

Combining (B7) and (B10), we can obtain the equilibrium values of λ and V , and I' can be determined from the inverse of V . Using (B1) and (B2) the equilibrium values of M^* and M can be calculated, and finally from I' we can obtain $p' = p(I')$ and $q' = q(I')$.

Equations (B7) and (B10) are depicted as curve (1) and curve (2) in Figure 4, respectively. Note that for both curves, the largest value of λ is 1. For curve (1), when $\lambda = \lambda^*$, $V(\lambda) = r$, and for curve (2), the smallest value of λ it can take upon is $\underline{\lambda} = \mu / (V^* - r)$, since in equilibrium $V > r$.

We shall assume that $V(\lambda)$ is convex in λ and that curves (1) and (2) intersect only once, so that the equilibrium is unique given μ , α , and other exogenous parameters.

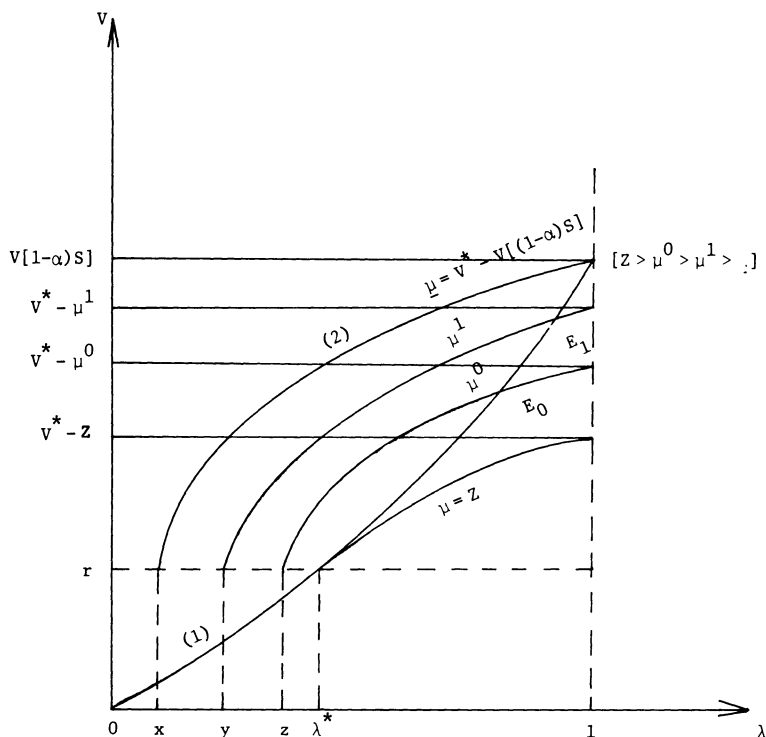
Comparative Statics Analysis with Respect to μ (for given α)—Figure 4

(1) *Effect of lower μ* : Notice that when μ becomes smaller, the equilibrium values of λ and V are larger. For example, when $\mu^0 \rightarrow \mu^1$, the equilibrium shifts from E_0 to E_1 . Therefore, when the search costs of the uninformed investors becomes smaller, there is a higher proportion of V^* firms, and also the (net) return offered by the inferior firms, V , is larger. As a result, the uninformed investors will be better off than when μ is large.

(2) *Characterization of equilibria by μ* : From Figure 4 when μ becomes smaller and smaller and attains the value $V^* - V[(1 - \alpha)S]$, curve (1) and (2) intersect at $\lambda = 1$ and $V = V[(1 - \alpha)S]$. $\mu < V^* - V[(1 - \alpha)S]$, as we have proved, is the necessary condition for the $\{(p^*, q^*)\}$ equilibrium. Now we have shown that it is also sufficient since when $\mu < V^* - V[(1 - \alpha)S]$, no other equilibria can exist.

Also, when μ becomes larger and larger and attains the value Z , where $Z = \lambda^*(V^* - r)$, the equilibrium is at $\lambda = \lambda^*$ and $V = r$. The lowest value for V in equilibrium is r , so when $\mu > Z$, we still have the equilibrium (λ^*, r) , but the equality condition (iii) (a) in Proposition 3.2 will be substituted by the inequality (iii) (b).

From the above analysis, the classification of equilibrium when there are some investors with zero cost (αN of them) and some with positive cost μ $((1 - \alpha)N$



(Note: $x \equiv \frac{V^* - V[(1-\alpha)S]}{V^* - r}$; $y \equiv \frac{\mu^1}{V^* - r}$;
 $z \equiv \frac{\mu^0}{V^* - r}$.)

Figure 4. Derivation of Competitive Intermediation Equilibria

of them) is as follows:

- (i) when $\mu < V^* - V[(1-\alpha)S]$, (p^*, q^*) only will be offered in equilibrium ($\lambda = 1$, $V = V^*$);
- (ii) when $V^* - V[(1-\alpha)S] \leq \mu < Z$, (p^*, q^*) and (p', q') will be offered in equilibrium ($\lambda^* < \lambda < 1$, $V > r$); and
- (iii) when $\mu \geq Z$, (p^*, q^*) and (p', q') will be offered ($\lambda = \lambda^*$, $V = r$).

When all investors have positive search costs, an equilibrium does not exist.

Characterization of Equilibrium by α (for given μ)

The effect of a decrease in α from α_1 to α_0 is illustrated in Figure 5; curve (1) shifts to the left and the equilibrium values of λ and V will decrease accordingly.

When α becomes smaller and smaller and attains the value

$$\alpha^* = \frac{1}{1 + \frac{V^* - r}{\mu} \frac{I_r}{S - I_r}}$$

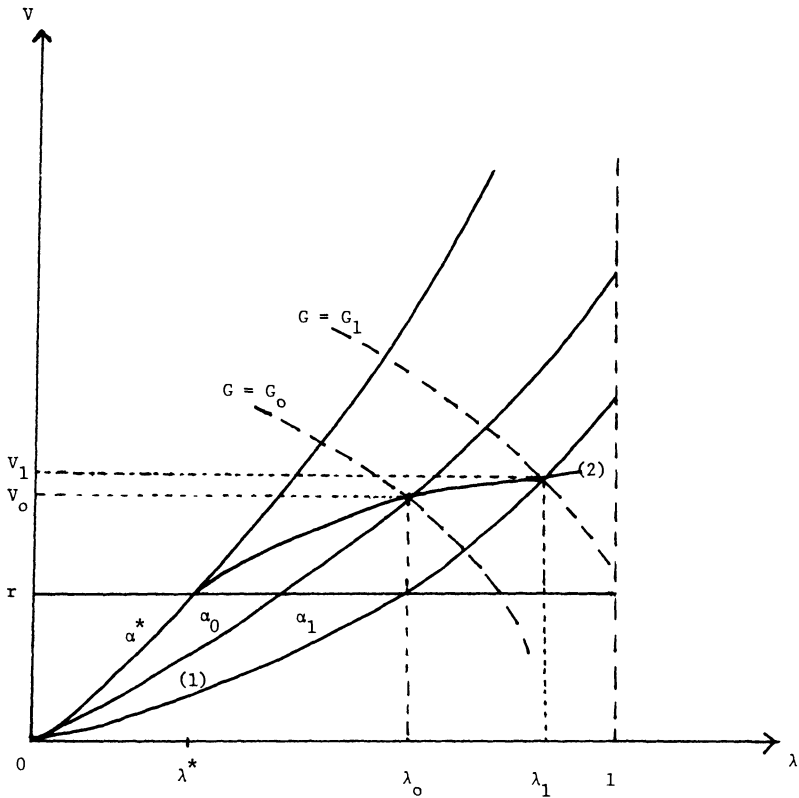


Figure 5. Comparative Statics Analysis of Competitive Intermediation Equilibria

curves (1) and (2) intersect at $\lambda = \underline{\lambda} = \mu / (V^* - r)$ and $V = r$. If α is smaller than α^* , we will have an equilibrium $(\lambda^*(\alpha), r)$, where $\lambda^*(\alpha)$ is as defined in (B9).

Proof of Proposition 4.3

Let $G(\lambda, V) = \lambda V^* + (1 - \lambda)V$. G measures the utility of all investors at the equilibrium (λ, V) . Notice when $\alpha_1 > \alpha_0$, $G_1 > G_0$. In a CI equilibrium, $\alpha = \gamma$, so that a high γ equilibrium will dominate a low γ equilibrium using the Pareto criterion.

Appendix C

Characterization of $F(\gamma)$

From Appendix B, when there are γN zero-cost informed investors and $(1 - \gamma)N$ high-cost μ investors (note that γ , the fraction of institutional holdings, is equivalent to α in Appendix B), the equilibrium distribution of returns as

characterized by γ , for given μ , is:

(i) for

$$1 \geq \gamma \geq \underline{\gamma} \equiv \left\{ 1 + \frac{(V^* - r)}{\mu} \frac{I_r}{(S - I_r)} \right\}^{-1}$$

the equilibrium distribution of returns is characterized by $(\lambda(\gamma), V(\gamma))$, with

$$\lambda(\gamma)(V^* - V(\gamma)) = \mu. \quad (C1)$$

(ii) for $0 \leq \gamma < \underline{\gamma}$, the equilibrium is $(\lambda^*(\gamma), r)$, that is, $V = r$, where

$$\lambda^* = \frac{\gamma}{1 - \gamma} \frac{I_r}{S - I_r}. \quad (C2)$$

In a CI equilibrium,

$$d(\gamma) = [1 - \lambda(\gamma)][V^* - V(\gamma)]. \quad (C3)$$

Therefore, when $1 \geq \gamma \geq \underline{\gamma}$,

$$F(\gamma) = \frac{\gamma d(\gamma)N}{M(\gamma)} = \frac{\gamma N}{\lambda(\gamma)M(\gamma)} [1 - \lambda(\gamma)]\mu. \quad (C4)$$

Market-clearing condition (refer to Proposition 3.2 or Equations (B1) and (B2) in Appendix B) implies

$$S - I(\gamma) = \frac{\gamma N}{\lambda(\gamma)M(\gamma)} \quad (C5)$$

and

$$I(\gamma) = \frac{(1 - \gamma)N}{M(\gamma)}. \quad (C6)$$

Thus,

$$F(\gamma) = [1 - \lambda(\gamma)][S - I(\gamma)]\mu. \quad (C7)$$

From Appendix B, for given μ , $\lambda(\gamma)$ and $I(\gamma)$ increases in γ . Therefore, $F(\gamma)$ is decreasing in γ for given μ when $1 \geq \gamma \geq \underline{\gamma}$.

When $0 \leq \gamma < \underline{\gamma}$, $V(\gamma) = r$ and $I(\gamma) = I_r$, so

$$M(\gamma) = [(1 - \gamma)N]/I_r. \quad (C8)$$

Using (C2), (C3), and (C8), for $0 \leq \gamma < \underline{\gamma}$,

$$F(\gamma) = (V^* - r)I_r \frac{\gamma}{1 - \gamma} \left[1 - \frac{\gamma}{1 - \gamma} \frac{I_r}{S - I_r} \right] \quad (C9)$$

Taking the derivative of $F(\gamma)$ with respect to γ , we find that

$$F'(\gamma) \geq 0 \quad \text{iff} \quad \gamma \leq (S - I_r)/(S + I_r) \quad (C10)$$

Comparing γ with $(S - I_r)/(S + I_r)$, we have

$$\gamma \geq (S - I_r)/(S + I_r) \quad \text{iff} \quad \mu \geq (V^* - r)/2 \quad (\text{C11})$$

Let $F^* = \max\{F(\gamma), 0 \leq \gamma \leq 1\}$, and γ_m be the maximizer of F in the interval $(0, 1)$, that is, $F^* = F(\gamma_m)$. We have

(i) when $\mu > (V^* - r)/2$, $\gamma_m = (S - I_r)/(S + I_r)$ and

$$F^* = F[(S - I_r)/(S + I_r)] = (V^* - r)(S - I_r)/4$$

and

(ii) when $\mu \leq (V^* - r)/2$, $\gamma_m = \underline{\gamma}$ and

$$F^* = F(\underline{\gamma}) = (S - I_r)[1 - (\mu/(V^* - r))]\mu.$$

Let $F_1^* \equiv F[(S - I_r)/(S + I_r)]$ and $F_2^*(\mu) \equiv F(\underline{\gamma}(\mu))$, we can show that F_2^* is concave and increasing in μ for $\mu \leq (V^* - r)/2$, and $F_1^* \geq F_2^*(\mu)$ for all μ in the same domain.

Therefore F^* as a function of μ is as shown in Figure 2.